Madi (Madeline) Abio

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Education

Griffith University — Gold Coast, Queensland
Bachelor's of Computer Science / Bachelor's of Electrical Engineering (Honours) — *Mar 2022 — Nov 2026*

- Cumulative GPA: 6.32/7

University of Utah — Salt Lake City, Utah, USA
International Trimester Exchange — *2024*

Awards & Scholarships

- **Academic Excellence** | *Griffith University, 2025*
- **Academic Excellence** | *Griffith University, 2024*
- **Academic Excellence** | *Griffith University, 2023*
- **Chancellor's Scholarship** | *Griffith University, 2025*
- **Amplify Connector** | *Energy Queensland, 2025 – Present*
- **Future of Energy Scholarship** | *Energy Queensland, 2023 – 2025*
- **Students in Power Bursary** | *Australian Institute of Power, 2022 – 2023*
- **Brighter Futures Scholarship** | *The Abedian Foundation, 2022*

Work Experience

Founding Engineer — Aug 2025 — Present
VPTech — *Gold Coast, QLD*

- Own end-to-end feature delivery across web applications, backend services, AI agents and cloud infrastructure for an AI-native clinical platform.
- Built the company's first AI agent, a proof-of-concept that validated the product's flagship feature and set the architectural direction for the agent platform.
- Own code quality and maintainability, building and maintaining automated CI checks covering linting, security scanning, architectural boundaries and design-system consistency.
- Mentor engineers in code cleanliness and engineering best practices through code review and team-wide standards.
- Drive technology discovery by scouting and evaluating emerging tools for adoption.
- Lead sprint planning.
- Improved developer experience by introducing stacked PRs, on-demand preview environments and observability tooling for debugging agent behaviour.

Grid Technology Undergraduate Engineer — Jan 2025 — Sept 2025
Energy Queensland — *Brisbane, QLD*

- Developed an anomaly detection machine learning model to detect neutral faults targeting deployment across 10,000 smart meters (see: Neutral Fault Detection project).
- Streamlined daily operational data analysis by building a solar panel data dashboard in Python/Power BI, reducing manual processing time from 15 minutes to 15 seconds.
- Managed project delivery using Agile principles, including facilitating weekly sprints.
- Automated various non-technical workflows.
- Coordinated two internal networking events for interns and graduates.

Undergraduate Research Assistant — Mar 2024 — Jun 2024
Griffith University — *Gold Coast, QLD*

- Performed data pre-processing for a machine learning research project by cleaning and structuring large datasets using Pandas (Python), Bash and the HDF5 format.

Cyber Security Platforms Intern — Nov 2023 — Feb 2024
Energy Queensland — *Brisbane, QLD*

- Developed a strong foundation in computer networking through self-directed training and mentorship, covering the OSI model, network protocols, and enterprise network architecture.

- Resolved ServiceNow tickets by applying network and security protocol knowledge.
- Gained exposure to enterprise cybersecurity practices, including cloud security, firewall management (Panorama), and Netskope.

Projects

Full-Stack LeetCode & Codeforces Web App (madiab.io) — TypeScript, March 2026

- Designed a full-stack web app to host my LeetCode solve times. Extended this to also host my portfolio.
- NestJS backend, Next.js frontend, PostgreSQL database, Prisma ORM and Better Auth for authentication.
- Dynamically linked the projects section to my GitHub repositories via their public API.
- Hosted on Railway.

Digital Bass Synthesizer on Microcontroller — Embedded C, Sept 2025 – Oct 2025

- Designed a digital bass synthesizer running at a 48 kHz sample rate on the TM4C1294NCPDT microcontroller.
- Developed firmware in bare-metal embedded C to generate waveforms, control ADSR envelopes, and handle user input from potentiometers (via analog-to-digital converters) and matrix keypad.
- Engineered a custom external clock circuit to bridge the I^2C peripheral for compatibility with the I^2S DAC.
- Configured SPI peripheral to drive the 240x320 LCD for waveform and real-time ADSR value visualization.
- Implemented DMA ping-pong buffers for continuous audio streaming and optimized ISR timing for stable waveform output.

Neutral Fault Detection (Anomaly Detection) — Python, Feb 2025 – Sept 2025

- Developed an in-house neutral fault detection algorithm based on an internal research paper to investigate excessive false positives produced by a third-party solution that also lacked coverage across all smart meters.
- During implementation and side-by-side evaluation, identified systematic false positives in the third-party system on solar PV sites, caused by violated voltage–current modelling assumptions that made normal solar behaviour resemble neutral fault signatures.
- Proposed a revised approach that explicitly accounted for solar PV electrical behaviour.
- Designed and prototyped a scalable architecture targeting deployment across 10,000+ smart meters.
- Built a parallelized feature extraction and model evaluation pipeline, and applied time-ordered, meter-isolated validation with systematic error and failure-mode analysis.

Encryption Client — C++, May 2025

- Developed a terminal-based multi-round message encryption/decryption client, applying OOP principles with abstract base classes, inheritance, and polymorphism for modular, reusable design.
- Implemented a finite-state machine menu interface with input validation and exception handling, leveraging STL and lambda functions for efficient message processing.

Clubs & Societies

- **Ravi's Study Program** | *Mentee, 2026*
- **Griffith University Coding Club (GCC)** | *ICPC Div B Competitor & Member, 2025 – Present*
- **UQ Computing Society** | *Lead Guitar Player (UQCS Band), Competitive Programmer, 2025 – Present*
- **UQ Fintech Society** | *Member, 2025 – Present*
- **Griffith University Advanced Robotics Development (GUARD)** | *Member, 2023 – Present*
- **Griffith University Gold Coast IEEE Student Branch** | *Administrator, 2023*

Skills

- **Programming Languages**: Bash, C, C++, C#, Python, TypeScript
- **Software Development & Tools**: Agentic Software Development (with Claude Code), AWS, CI/CD, Drizzle, Fastify, Git, Linux, NestJS, Next.js, oRPC, PostgreSQL, Prisma, React
- **Machine Learning & Data Science**: Applied Machine Learning, Data Science, h5py, NumPy, Pandas, scikit-learn
- **LLMOps**: AgentCore, LangChain, LangGraph, LLM orchestration
- **Embedded & FPGA**: Embedded/Bare Metal C, PCB Design, SystemVerilog
- **Software Practices**: Object-Oriented Programming, Test-Driven Development
- **Professional Skills**: Communication, Problem Solving, Teamwork, Technical Writing